In [1]:
```
library(tidyverse)
install.packages('wooldridge')
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.
0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   3.5.1     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts
() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all co
nflicts to become errors
Installing package into '/srv/r'
(as 'lib' is unspecified)
```

In [2]:
```
options(scipen=999)
```

In [3]:
```
library("wooldridge")
data(beauty, package = "wooldridge")
head(beauty)
```

A data.frame: 6 × 1

| | wage | lwage | belavg | abvavg | exper | looks | union | goodhlth | black | fem |
|---|---|---|---|---|---|---|---|---|---|---|
| | <dbl> | <dbl> | <int> | <int> | <int> | <int> | <int> | <int> | <int> | <i |
| **1** | 5.73 | 1.745715 | 0 | 1 | 30 | 4 | 0 | 1 | 0 | |
| **2** | 4.28 | 1.453953 | 0 | 0 | 28 | 3 | 0 | 1 | 0 | |
| **3** | 7.96 | 2.074429 | 0 | 1 | 35 | 4 | 0 | 1 | 0 | |
| **4** | 11.57 | 2.448416 | 0 | 0 | 38 | 3 | 0 | 1 | 0 | |
| **5** | 11.42 | 2.435366 | 0 | 0 | 27 | 3 | 0 | 1 | 0 | |
| **6** | 3.91 | 1.363537 | 0 | 0 | 20 | 3 | 0 | 0 | 0 | |

In [4]:
```
female_beauty_data <- subset(beauty, female ==1)
```

In [5]:
```
reg1 <- lm(lwage ~ looks + educ + married + goodhlth, data = female_beauty_c
summary(reg1)
```

```
Call:
lm(formula = lwage ~ looks + educ + married + goodhlth, data = female_beauty
_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.28573 -0.31116  0.04698  0.30769  3.01283 

Coefficients:
             Estimate Std. Error t value            Pr(>|t|)    
(Intercept)  0.211115   0.165918   1.272               0.204    
looks        0.034753   0.032800   1.060               0.290    
educ         0.077044   0.009292   8.291 0.00000000000000144 ***
married     -0.043925   0.046761  -0.939               0.348    
goodhlth     0.032507   0.086081   0.378               0.706    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.485 on 431 degrees of freedom
Multiple R-squared:  0.1523,	Adjusted R-squared:  0.1444 
F-statistic: 19.36 on 4 and 431 DF,  p-value: 0.00000000000001169
```

In [6]:
```
female_beauty_data <- mutate(female_beauty_data, blackxlooks = black * looks
```

In [7]:
```
reg2 <- lm(lwage ~ looks + educ + married + goodhlth +  blackxlooks, data =
summary(reg2)
```

```
Call:
lm(formula = lwage ~ looks + educ + married + goodhlth + blackxlooks, 
    data = female_beauty_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.27281 -0.32371  0.04108  0.31739  2.81280 

Coefficients:
             Estimate Std. Error t value            Pr(>|t|)    
(Intercept)  0.164938   0.166159   0.993              0.3214    
looks        0.030573   0.032670   0.936              0.3499    
educ         0.079609   0.009304   8.556 <0.0000000000000002 ***
married     -0.035528   0.046642  -0.762              0.4466    
goodhlth     0.036566   0.085634   0.427              0.6696    
blackxlooks  0.054281   0.022779   2.383              0.0176 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4824 on 430 degrees of freedom
Multiple R-squared:  0.1633,	Adjusted R-squared:  0.1536 
F-statistic: 16.79 on 5 and 430 DF,  p-value: 0.000000000000003659
```

In [8]:
```
mean(female_beauty_data$looks[female_beauty_data$black==1], na.rm = TRUE)
```

3.12

In [9]:
```
mean(female_beauty_data$looks[female_beauty_data$black==0], na.rm = TRUE)
```

3.21243523316062

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [15]:
```
mean(female_beauty_data$wage[female_beauty_data$black==1 & female_beauty_dat
```

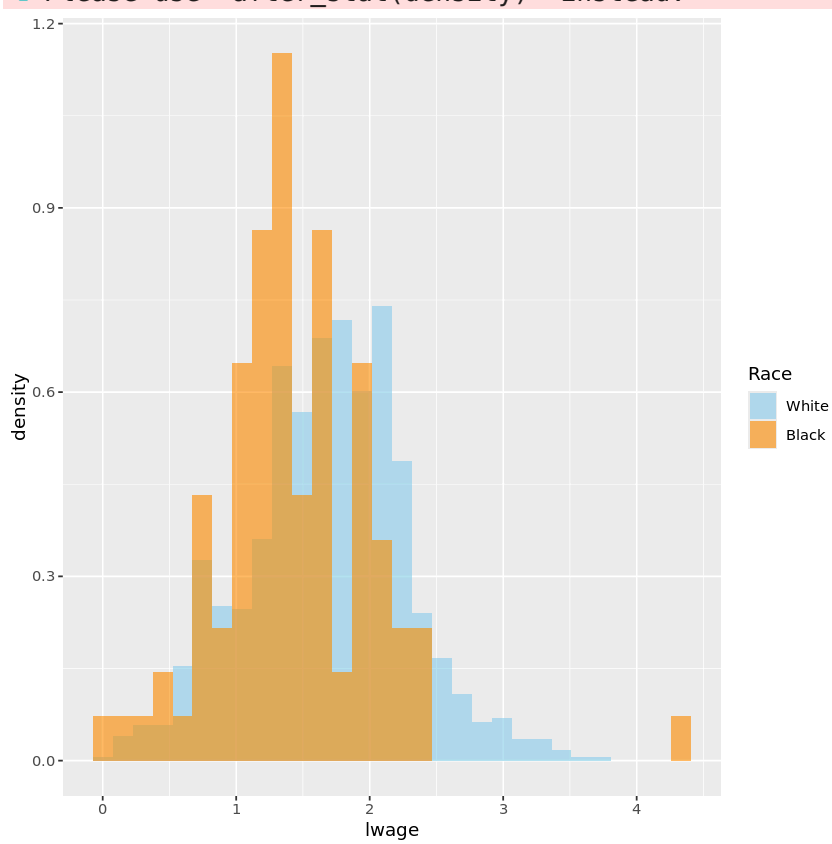
7.21000003814697

In [16]:
```
mean(female_beauty_data$wage[female_beauty_data$black==0 & female_beauty_dat
```

6.0237179352687

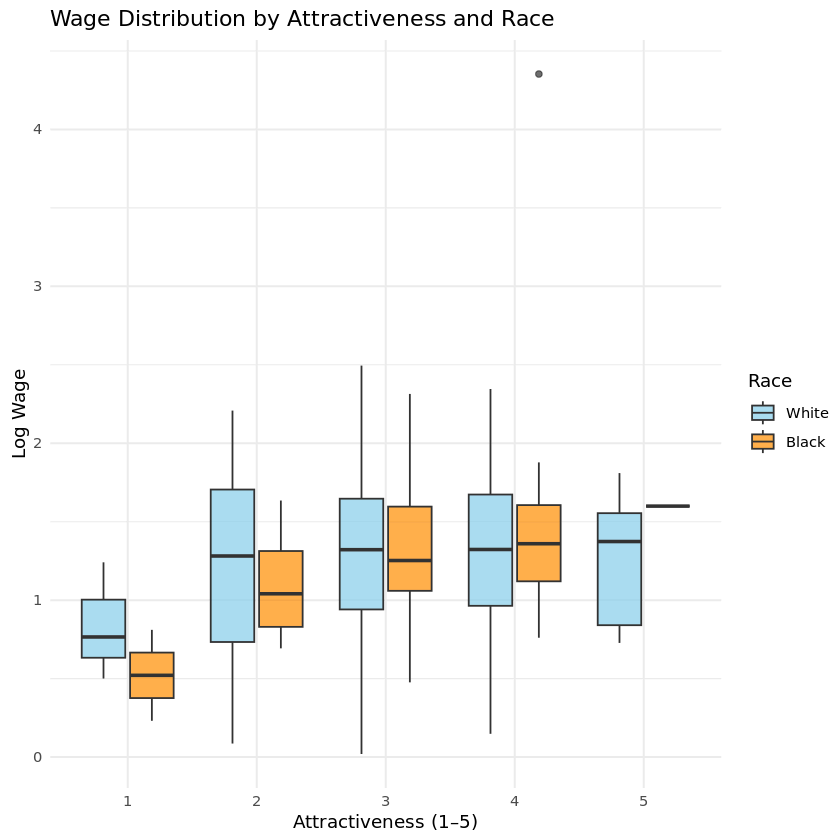
In [18]:
```
ggplot(beauty, aes(x = lwage, fill = factor(black))) +
  geom_histogram(aes(y= ..density..), position = "identity", alpha = 0.6, bi
  scale_fill_manual(values = c("skyblue", "darkorange"),
                    labels = c("White", "Black"),
                    name = "Race")
```

```
Warning message:
“The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
ℹ Please use `after_stat(density)` instead.”
```



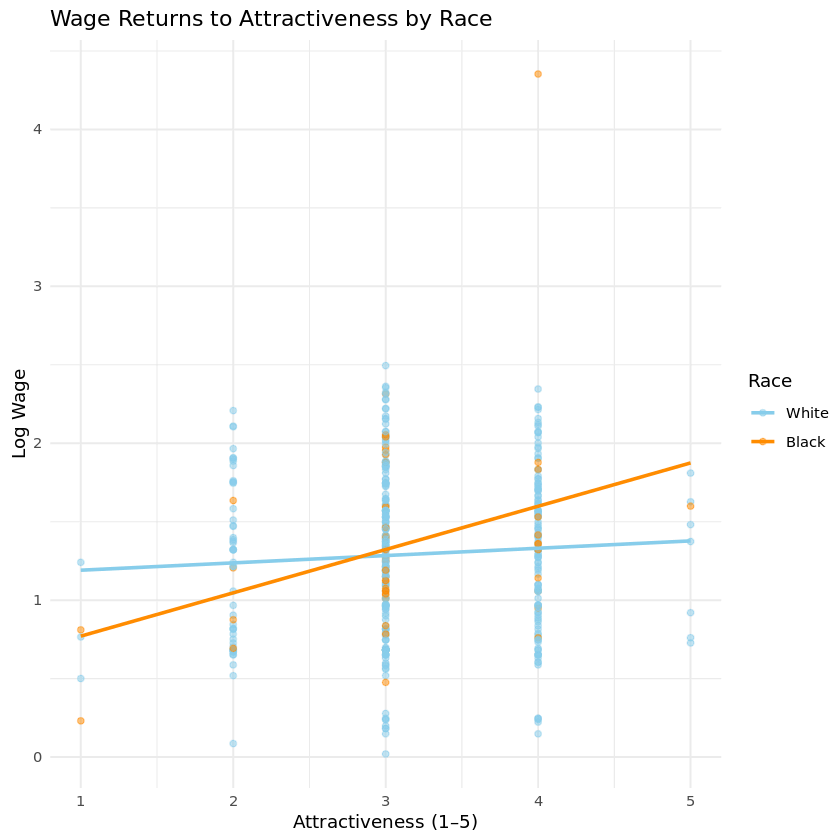

In [19]:
```
ggplot(beauty %>% filter(female == 1), aes(x = factor(looks), y = lwage, fil
  geom_boxplot(alpha = 0.7) +
```

```
  scale_fill_manual(values = c("skyblue", "darkorange"),
                    labels = c("White", "Black"),
                    name = "Race") +
  labs(x = "Attractiveness (1–5)", y = "Log Wage",
       title = "Wage Distribution by Attractiveness and Race") +
  theme_minimal()
```



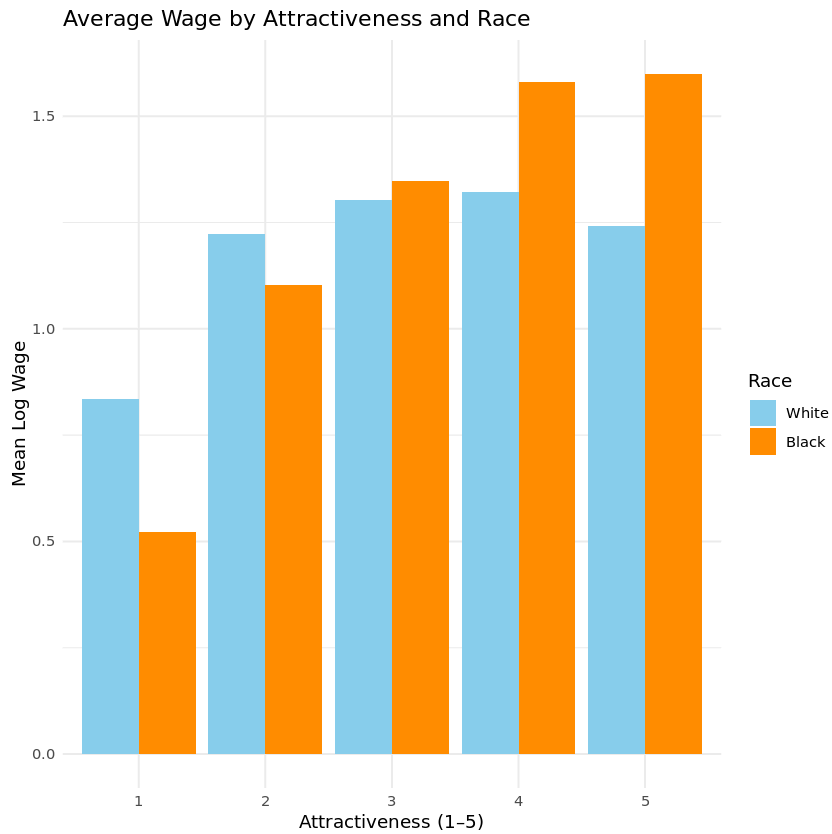

In [20]:
```
ggplot(beauty %>% filter(female == 1), aes(x = looks, y = lwage, color = fac
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_color_manual(values = c("skyblue", "darkorange"),
                     labels = c("White", "Black"),
                     name = "Race") +
  labs(x = "Attractiveness (1–5)", y = "Log Wage",
       title = "Wage Returns to Attractiveness by Race") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

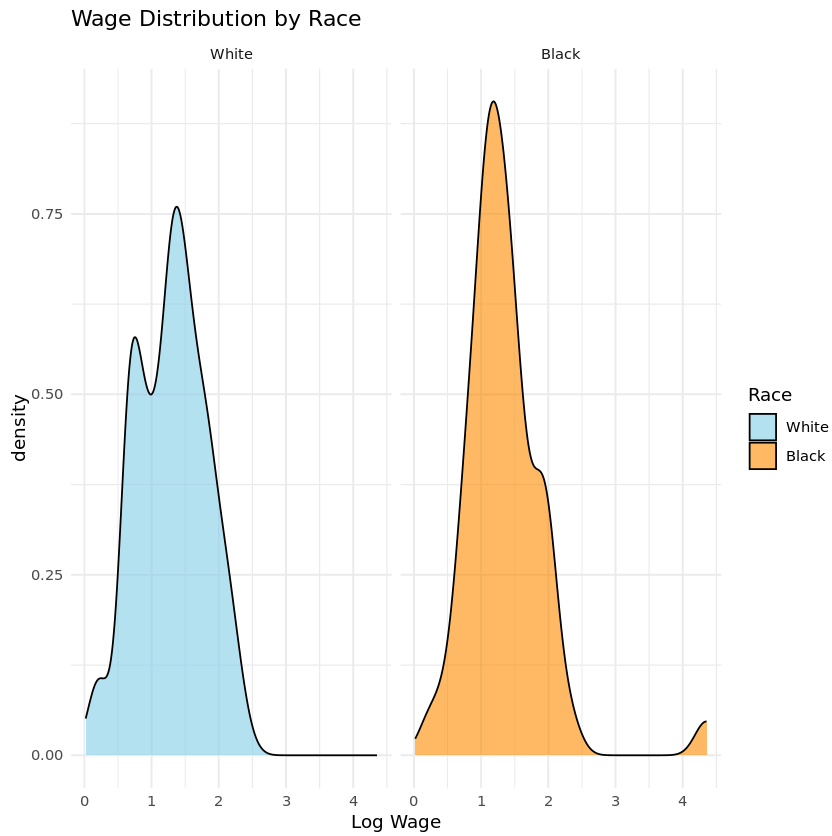



In [21]:
```
beauty %>%
  filter(female == 1) %>%
  group_by(looks, black) %>%
  summarise(mean_lwage = mean(lwage, na.rm = TRUE)) %>%
  ggplot(aes(x = factor(looks), y = mean_lwage, fill = factor(black))) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("skyblue", "darkorange"),
                    labels = c("White", "Black"),
                    name = "Race") +
  labs(x = "Attractiveness (1–5)", y = "Mean Log Wage",
       title = "Average Wage by Attractiveness and Race") +
  theme_minimal()
```

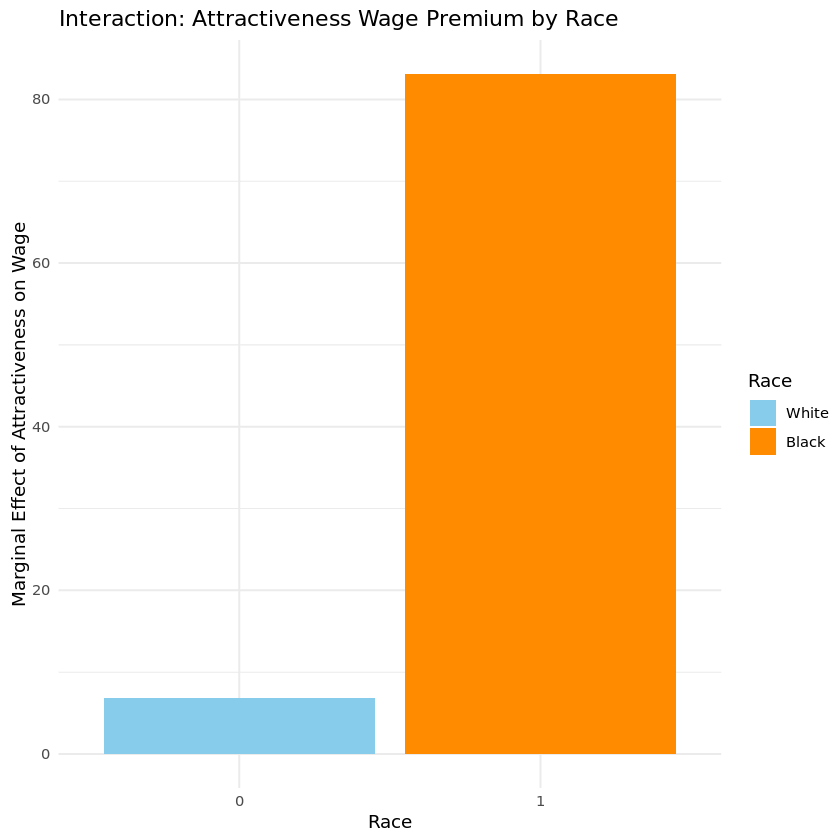
```
`summarise()` has grouped output by 'looks'. You can override using the
`.groups` argument.
```




In [22]:
```
ggplot(beauty %>% filter(female == 1), aes(x = lwage, fill = factor(black)))
  geom_density(alpha = 0.6) +
  scale_fill_manual(values = c("skyblue", "darkorange"),
                    labels = c("White", "Black"),
                    name = "Race") +
  facet_wrap(~factor(black), labeller = as_labeller(c("0" = "White", "1" = "
  labs(x = "Log Wage", title = "Wage Distribution by Race") +
  theme_minimal()
```




In [24]:
```
library(margins)
model <- lm(lwage ~ looks * factor(black) + educ + married + goodhlth,
            data = beauty %>% filter(female == 1))
margins_model <- margins(model, variables = "looks", at = list(black = c(0,
ggplot(margins_model, aes(x = factor(black), y = dydx_looks, fill = factor(b
  geom_col() +
  scale_fill_manual(values = c("skyblue", "darkorange"),
                    labels = c("White", "Black"),
                    name = "Race") +
  labs(x = "Race", y = "Marginal Effect of Attractiveness on Wage",
       title = "Interaction: Attractiveness Wage Premium by Race") +
  theme_minimal()
```




In [25]:
```
sum(beauty$female == 1 & beauty$black == 1, na.rm = TRUE)
```

50

In [26]:
```
sum(beauty$female == 1 & beauty$black == 0, na.rm = TRUE)
```

386

In [27]:
```
library(dplyr)

beauty %>%
  filter(female == 1) %>%  # Subset to women only
  count(black) %>%         # Count by race (black = 1/0)
  mutate(race = ifelse(black == 1, "Black Women", "White Women")) %>%
  select(race, n)
```

A data.frame: 2 × 2

| race | n |
|---|---|
| <chr> | <int> |
| White Women | 386 |
| Black Women | 50 |

In [30]:
```
beauty %>%
  filter(female == 1, looks >= 1) %>%  # Women with attractiveness >= 3
  count(black)
```

A data.frame: 2 × 2

| black | n |
|---|---|
| <int> | <int> |
| 0 | 386 |
| 1 | 50 |

In [ ]: